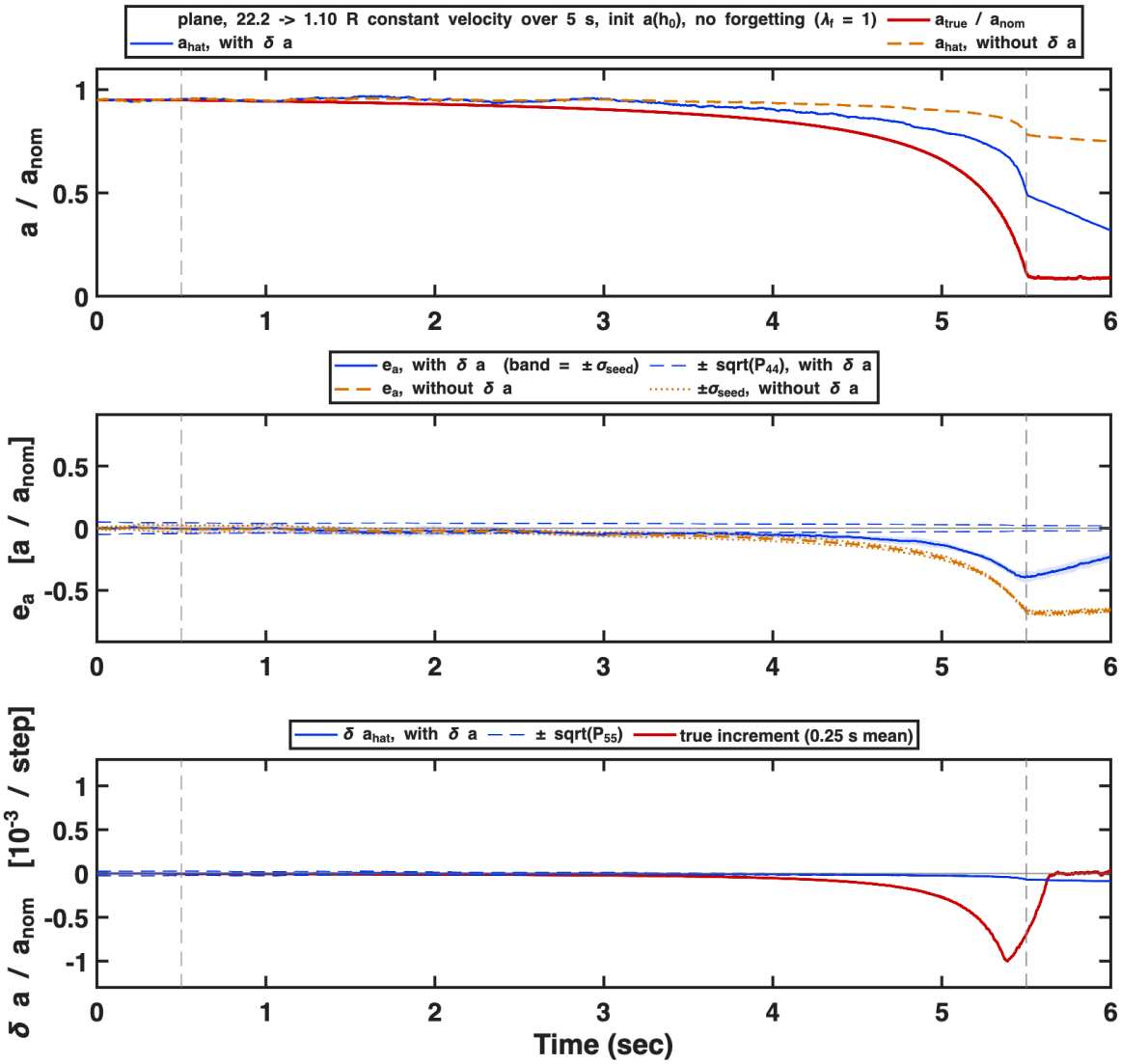
2026-09-24 report

Trajectory: hold 0.5 s, constant-velocity descent $22.2 \to 1.10\,R$ over T_d, hold 0.5 s. Plane. $\lambda_f = 1.$ 5 seeds.

1 First-order vs second-order gain model, $T_d = 5$ s

first order: $a[k+1] = a[k]$

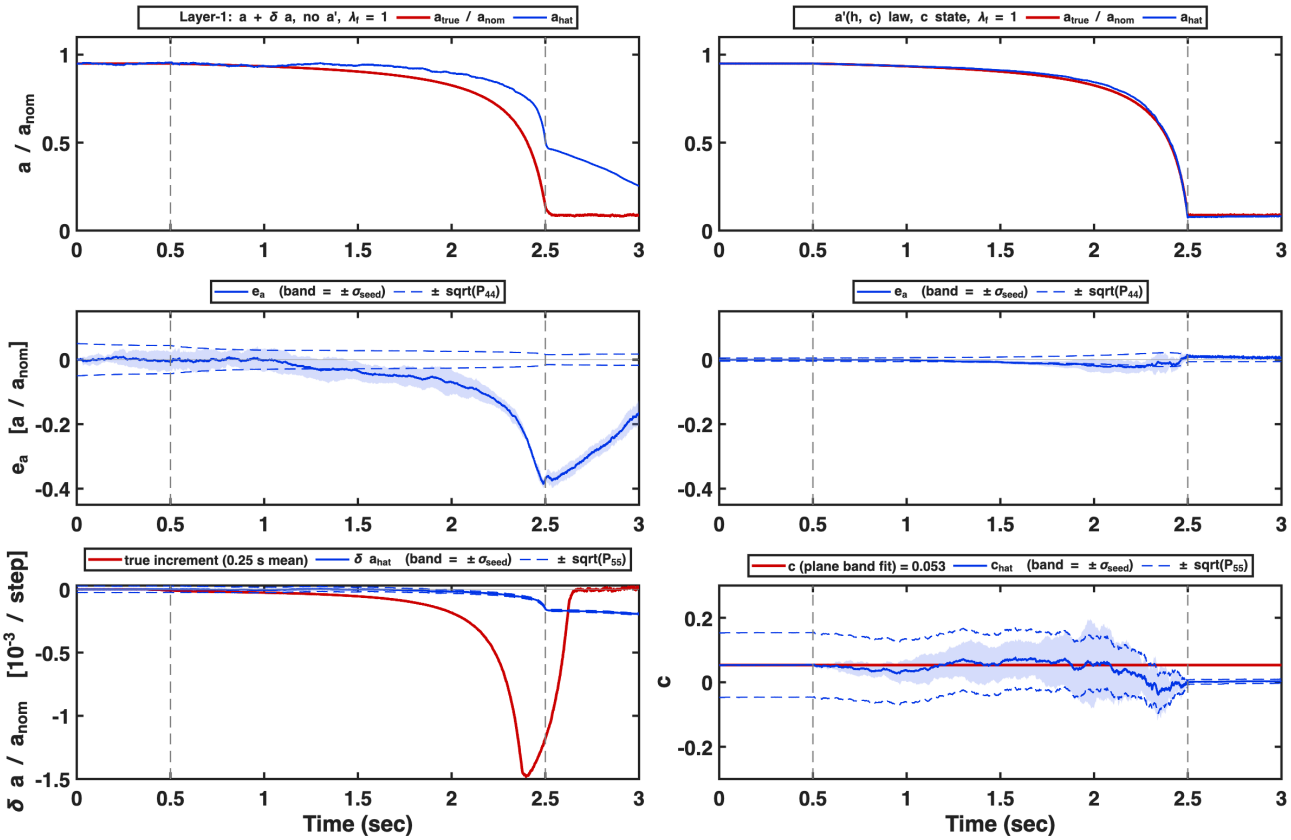
second order: $a[k+1] = a[k] + \delta a[k], \qquad \delta a[k+1] = \delta a[k]$



2 Second-order gain model vs slope law $a'(h, c)$, $T_d = 2$ s

second order: $a[k+1] = a[k] + \delta a[k], \qquad \delta a[k+1] = \delta a[k]$

slope law: $a[k+1] = a[k] + a'(h_d[k], c)\,\Delta h_d[k], \quad a' = \dfrac{1-c}{h^2}, \qquad c[k+1] = c[k]$



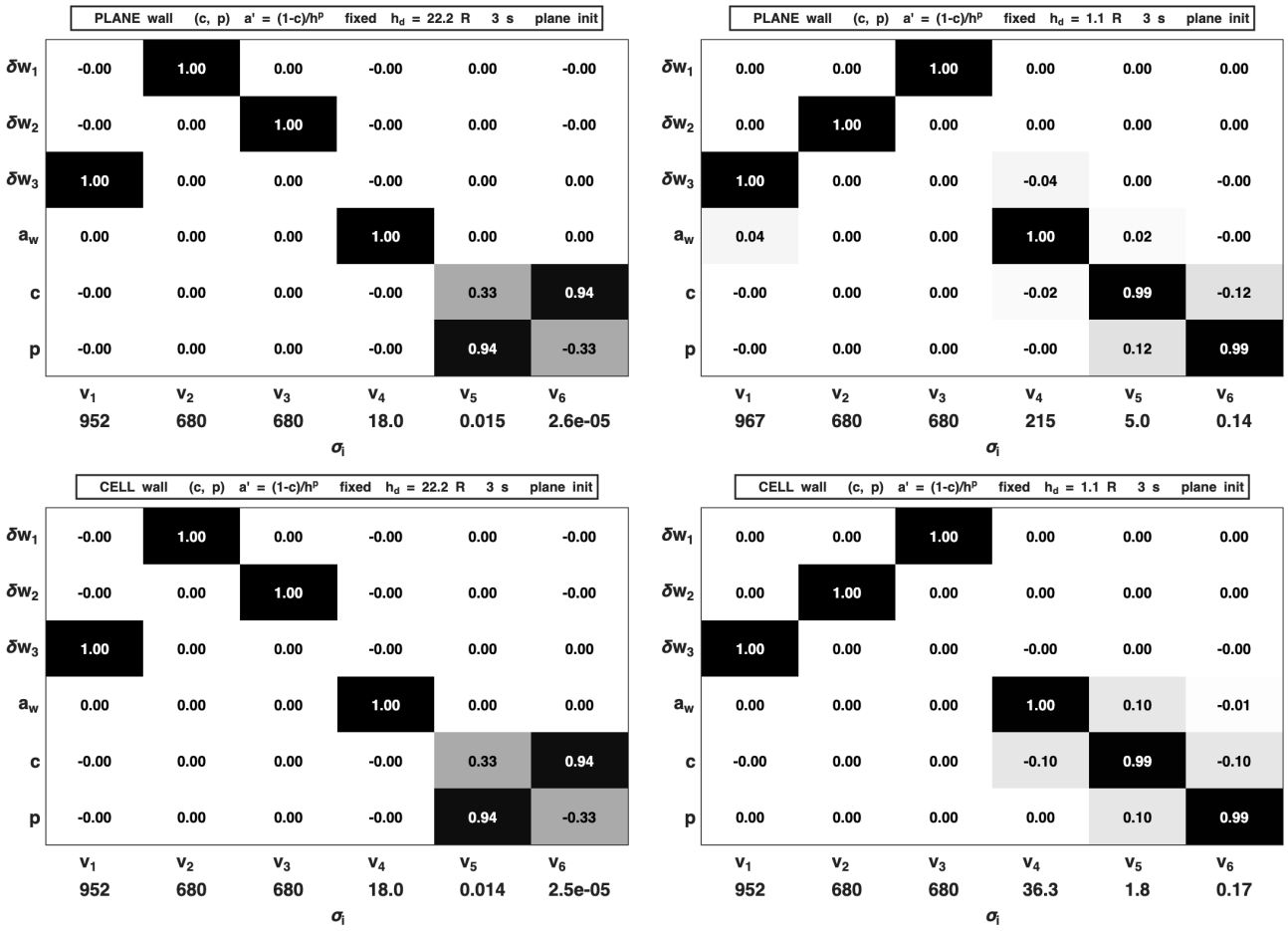
3 Slope law $a'(h;\, c, p)$, both constants estimated: fixed heights $22.2\,R$ and $1.10\,R$

Plane and cell wall, plane init, held 3 s, one seed. σ_i, v_i: singular values and directions of the observability matrix over the run.

$$a[k+1] = a[k] + a'(h_d[k];\, c, p)\,\Delta h_d[k], \qquad a' = \frac{1-c}{h^p}$$

$$c[k+1] = c[k]$$

$$p[k+1] = p[k]$$



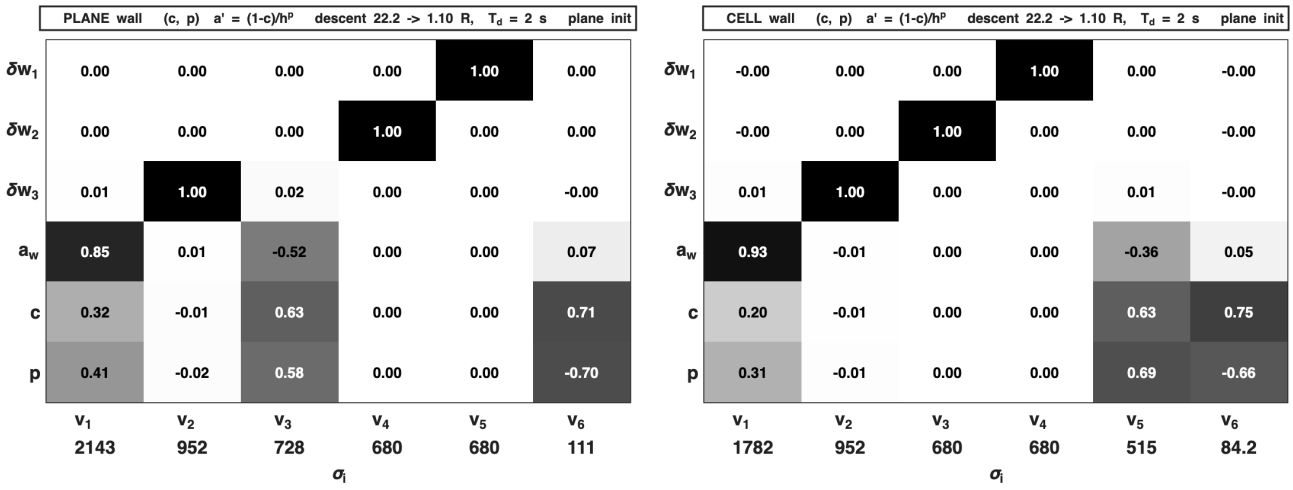
4 Slope law $a'(h;\, c, p)$, both constants estimated, $T_d = 2$ s: observability over the descent

Plane and cell wall, plane init, one seed.

$$a[k+1] = a[k] + a'(h_d[k];\, c, p)\, \Delta h_d[k], \qquad a' = \frac{1-c}{h^p}$$

$$c[k+1] = c[k]$$

$$p[k+1] = p[k]$$



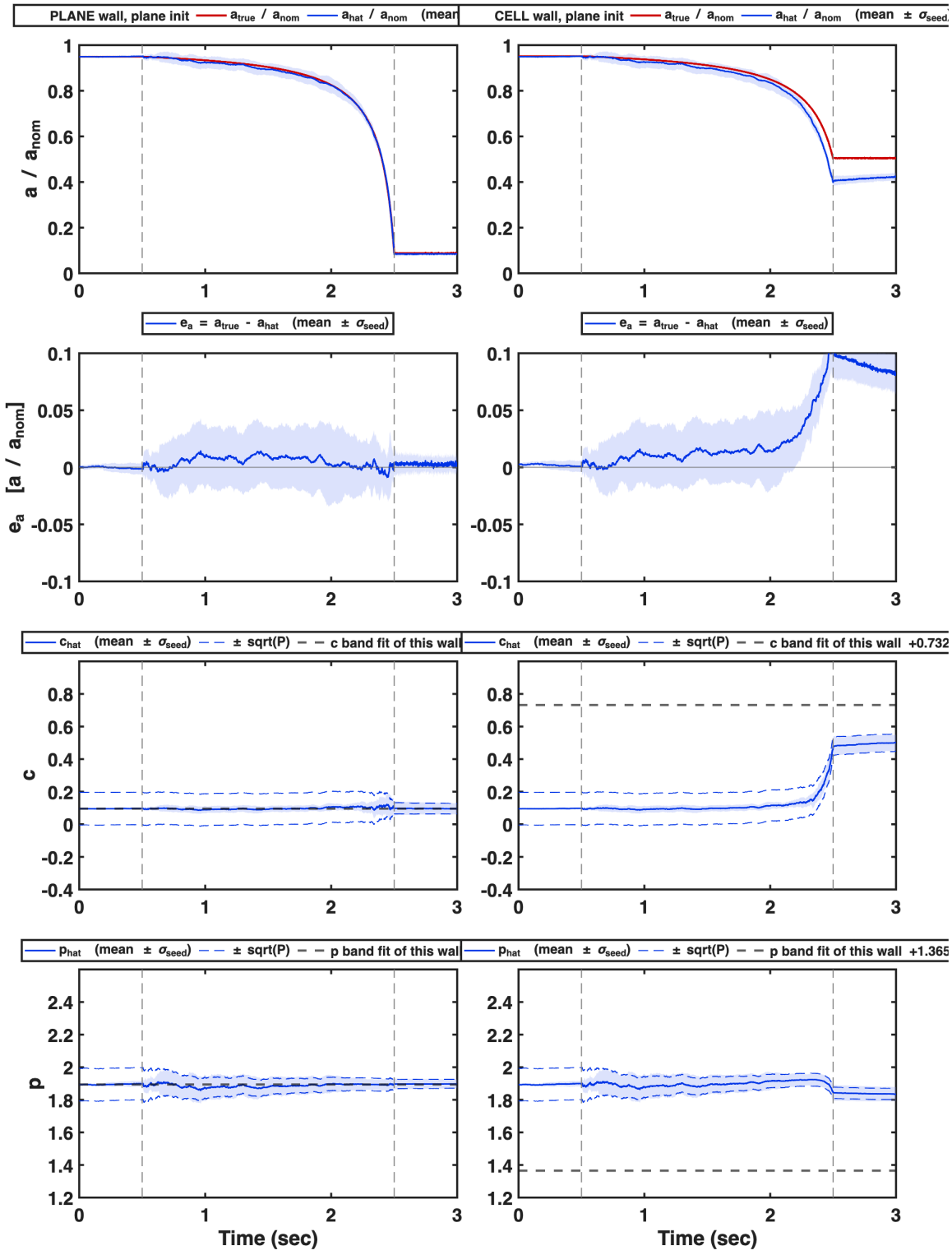
5 Slope law $a'(h;\, c, p)$, both constants estimated, $T_d = 2$ s: plane and cell wall

Plane init, 10 seeds.

$$a[k+1] = a[k] + a'(h_d[k];\, c, p)\, \Delta h_d[k], \qquad a' = \frac{1-c}{h^p}$$

$$c[k+1] = c[k]$$

$$p[k+1] = p[k]$$

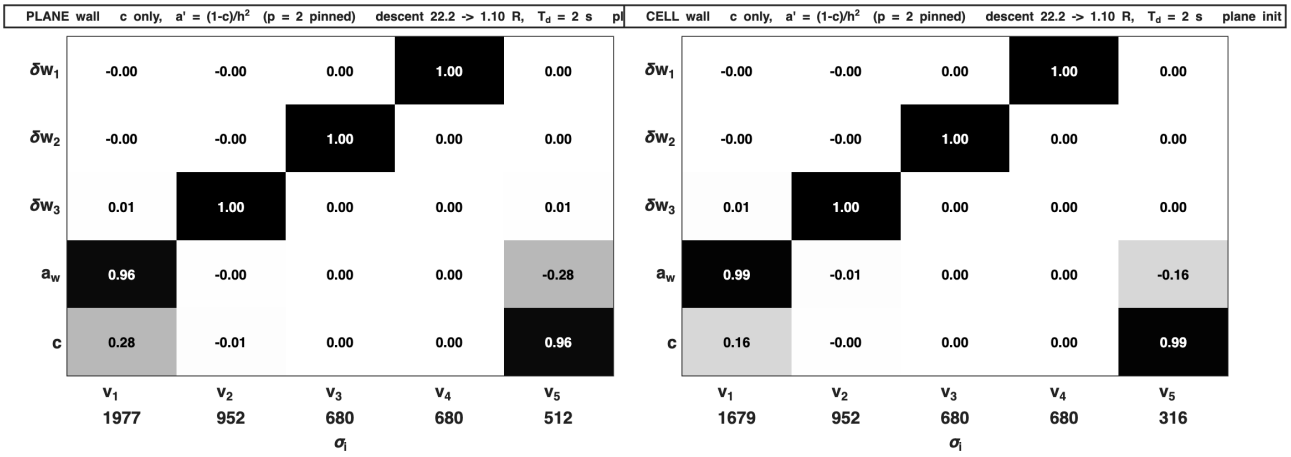


6 Slope law $a'(h, c)$, $p = 2$, one constant, $T_d = 2$ s: observability over the descent

Plane and cell wall, plane init $c = 0.053$, one seed. σ_i, v_i as in Section 3; p known, so it is not a state.

$$a[k+1] = a[k] + a'(h_d[k], c)\,\Delta h_d[k], \qquad a' = \frac{1-c}{h^2}$$

$$c[k+1] = c[k]$$

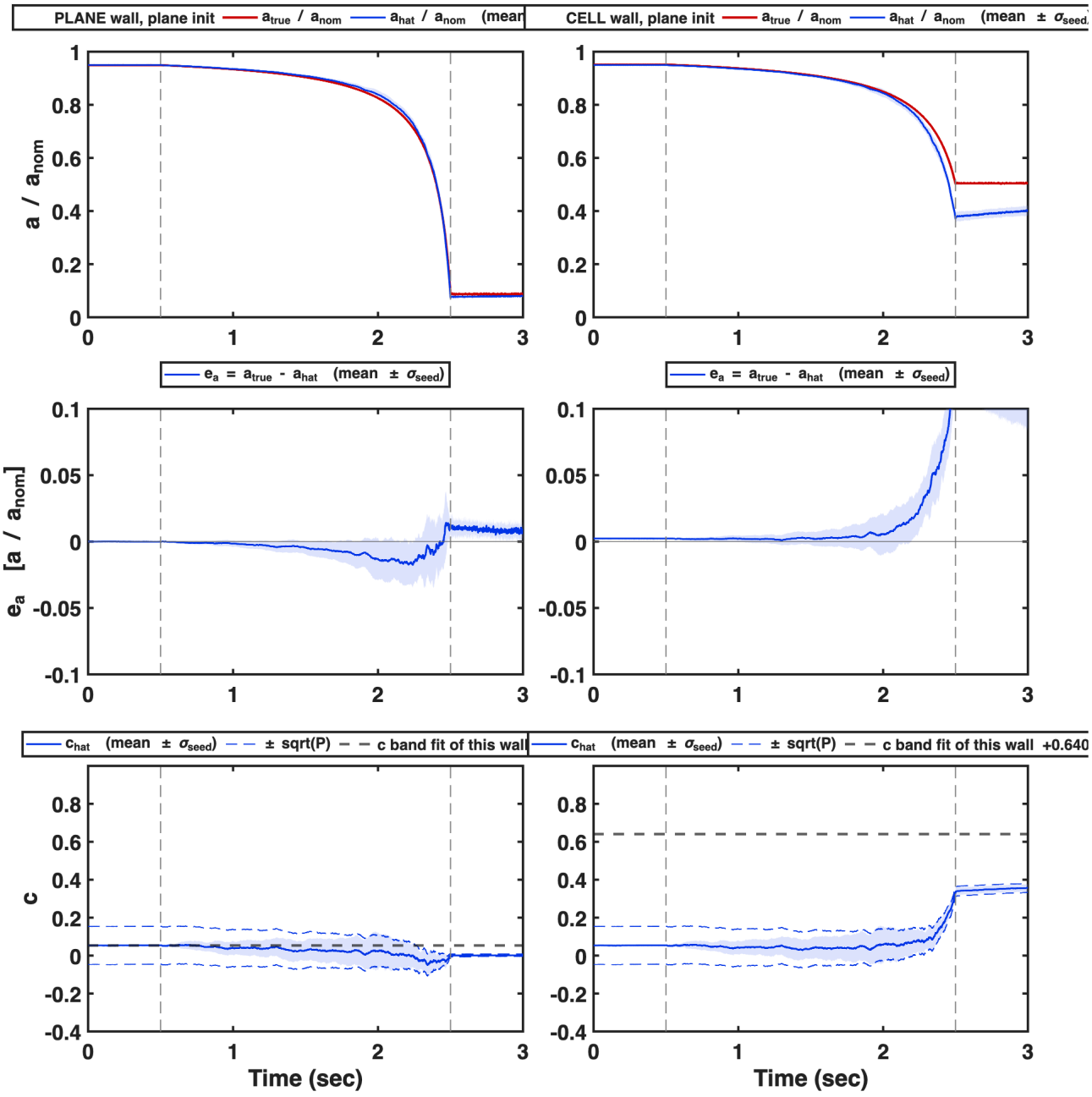


7 Slope law $a'(h, c)$, $p = 2$, one constant, $T_d = 2$ s: plane and cell wall

Plane init $c = 0.053$ (plane band fit at $p = 2$), prior 0.10, 10 seeds, $\lambda_f = 1$.

$$a[k+1] = a[k] + a'(h_d[k], c)\,\Delta h_d[k], \qquad a' = \frac{1-c}{h^2}$$

$$c[k+1] = c[k]$$



8 Covariance recursion: the update only shrinks P; where the forgetting factor enters

Kalman filter, one step (Meng 2025 (19)–(21)); c a constant state, $Q_{cc} = 0$. No measurement value appears in (19)–(21).

$$L = P^f H^T \left(H P^f H^T + R \right)^{-1}$$

$$P = (I - LH)\, P^f \;=\; P^f - P^f H^T \left(H P^f H^T + R \right)^{-1} H P^f \;\leq\; P^f$$

$$P^f[k+1] = F_e\, P\, F_e^T + Q, \qquad Q_{cc} = 0 \;\Rightarrow\; P^f_{cc}[k+1] = P_{cc}[k]$$

The same step for the c element, written in information ($1/P_{cc}$):

$$s_k = 1 - \frac{P_{cc}[k]}{P^f_{cc}[k]} \qquad \text{(fraction of } c\text{'s variance removed by this update; set by } P^f, H, R \text{ only)}$$

$$\frac{1}{P_{cc}[k]} = \frac{1}{P_{cc}[k-1]} + g_k, \qquad g_k = \frac{s_k}{(1 - s_k)\, P^f_{cc}[k]}$$

Forgetting on the c element only (after the update, before the next prediction):

$$P_{cc} \leftarrow P_{cc}/\lambda_k, \quad P_{cj} \leftarrow P_{cj}/\sqrt{\lambda_k} \qquad \Leftrightarrow \qquad \frac{1}{P_{cc}[k]} = \lambda_k \left(\frac{1}{P_{cc}[k-1]} + g_k \right)$$

$$\lambda_k = 1 \text{ (constant)}, \qquad \lambda_k = \lambda_f \text{ (uniform)}, \qquad \lambda_k = \exp(-\beta\, s_k) \text{ (information-driven)}$$

$$\lambda_k = \exp(-\beta\, s_k) \;\Rightarrow\; (1 - \lambda_k) \left(\frac{1}{P_{cc}[k-1]} + g_k \right) \approx \beta\, g_k$$

9 Slope law $a'(h, c)$, $p = 2$, one constant, $T_d = 2$ s: cell wall, three forgetting factors on c

Plane init $c = 0.053$, prior 0.10, 10 seeds. Forgetting on the c element after each update (Section 8), one choice per column.

$$a[k+1] = a[k] + a'(h_d[k], c)\,\Delta h_d[k], \qquad a' = \frac{1-c}{h^2}$$

$$c[k+1] = c[k]$$

$$P_{cc} \leftarrow P_{cc}/\lambda_k, \qquad P_{cj} \leftarrow P_{cj}/\sqrt{\lambda_k}$$

left: $\lambda_k = \max\{\exp(-3\,s_k),\ 0.99\}, \qquad s_k = 1 - \dfrac{P_{cc}[k]}{P^f_{cc}[k]}$

middle: $\lambda_k = 0.999$

right: $\lambda_k = 1$

